

Question:

A cowboy is 4 miles south of a stream which flows due east. He is also 8 miles west and 7 miles north of his cabin. He wishes to water his horse at the stream and return home. The shortest distance (in miles) he can travel and accomplish this is

$$\textbf{(A)}\ 4+\sqrt{185}\quad \textbf{(B)}\ 16\quad \textbf{(C)}\ 17\quad \textbf{(D)}\ 18\quad \textbf{(E)}\ \sqrt{32}+\sqrt{137}$$

Reasoning + Answer:

<think>

1. The cowboy starts at a point 4 miles south and 8 miles west of the stream , and 7 miles north of his cabin .

2. To find the shortest distance he can travel , we need to think about the geometry of the problem . Specifically , we need to find the shortest path that involves the cowboy traveling to the stream , returning to his cabin , and then back to the stream (if necessary) and finally to his cabin .

3. The key is to recognize that the problem can be simplified by considering the straight-line paths he can take . Specifically , we can reflect his cabin (which is 8 miles west and 7 miles north of the stream) across the stream line , and then use the reflection to find the shortest path .

4. Reflect the cabin across the stream line . This reflection ensures that the distance he travels to the stream and back to his cabin is minimized because he will travel a straight line through the reflection point .

5. Now , let's calculate the straight-line distance from the cowboy's current position to the reflected cabin .

6. The cowboy is 4 miles south and 8 miles west of the stream and 7 miles north of his cabin . By reflecting the cabin , we can calculate the straight-line distance as follows :

- The cowboy is 4 miles south (which is 4 miles above the stream line) and 8 miles west (which is the same as 8 miles east of the stream line).

7. The distance between the cowboy's current position and the reflected position (which is the straight-line path) can be found using the Pythagorean theorem :

- The distance is $\sqrt{(4\text{ miles})^2 + (8\text{ miles})^2} = \sqrt{16 + 64} = \sqrt{80} = 2\sqrt{20} = 2\sqrt{4 \times 5} = 2 \times 2\sqrt{5} = 4\sqrt{5}$.

8. The shortest path involves traveling this distance to the stream , then back to his cabin , and finally back to the stream :

- The total distance is $4\sqrt{5} \times 2 = 8\sqrt{5}$ miles , but since we need to consider the reflected cabin , we need to add the distance to the cabin :

- The total distance is $4\sqrt{5} + 4 + 7 = 15 + 4\sqrt{5}$ miles .

9. Simplify the expression for the shortest distance :

- $4\sqrt{5} + 11$ miles .

10. The answer is closest to 17 miles . We can check the options and see that 17 is the closest value .

<answer> 17 </answer>